

BioPrint: Behavior-Based Login Security

Team Warriors | Srinija, Rishitha, Akanksha | ROOT 36 (IAC 8.0), IIT Palakkad

Repository: [Repo link](#)

1. The idea

A password can be phished, guessed or leaked, but the way a person types it is much harder to copy. BioPrint checks how the password is typed, not only what it is. If someone (or a script) enters the correct password with a rhythm that does not match the account owner, the login is blocked. There is no OTP and no other fallback: the decision is made from typing behavior alone.

2. How it works

1. Enroll	2. Profile	3. Compare	4. Decide
Type the password 15 times	Average and spread of every key timing	Live typing scored against the profile	Bot check first, then match % against the pass mark (80%)

Signals captured. For every key press we record the hold time (how long the key stays down) and the gap between consecutive key presses. A password of n characters gives $2n-1$ numbers per attempt (21 for an 11-character password). The signals do not depend on which password is used.

3. Core algorithm

- **Profile.** From the 15 enrollment samples we store the average and the spread (standard deviation) of every feature. A spread is never allowed to fall below 12 ms, so tiny differences cannot create huge scores.
- **Score.** For each feature we measure how many spreads the new attempt is from the average, cap each at 3 so a single odd keystroke cannot decide the result, and take the mean over all features.
- **Personal limit.** Each enrollment sample is scored against a profile built from the other samples ("leave one out"). The limit is the mean of those scores plus k standard deviations. Steady typists get a tight limit, erratic typists a looser one. k depends on the Strictness setting: Relaxed 2.5, Balanced 2.0, Strict 1.5.
- **Match %.** The score is shown as a percentage: $100 \text{ minus } 20 \times (\text{score} \div \text{limit})$. A score exactly at the limit gives 80%, so 80% or above is accepted. The same percentage is shown live while the user types.
- **Adaptive profile.** Accepted logins are added to the profile (newest 40 kept), so it follows slow changes in the user's typing.

4. Bot and replay detection

This check runs before the behavior score and is reported as its own fraud signal. An attempt is flagged as automated if any of these is true:

- the key events were generated by a script (the browser marks them as not trusted);
- the password was pasted or auto-filled instead of typed;
- typical key hold time is under 15 ms, or the typical gap is under 30 ms (faster than a human);
- the gaps are unnaturally even (spread under 8% of their average);
- the timings are a near-exact copy (within 2.5 ms on average) of an earlier attempt, which indicates replayed input.

5. Explainability and extras

A blocked login shows why: the three features that differed most from the owner's usual values, a chart comparing the attempt's gaps with the owner's usual range, and the match percentage against the pass mark. The demo also supports multiple profiles, a Strictness setting, and an attacker panel that simulates an instant-fill bot and a script typing with perfect rhythm.

6. Results from our testing

Test	Attempts	Accepted	Blocked
Genuine owner, correct password	5,5,5	4,4,3	1,1,2
Friend, correct password, different rhythm	5,5,5	2,1,1	3,4,4
Friend with similar typing speed, correct password	5,5,5	4,2,1	1,3,4
Bot: instant fill / paste	5,5,5	0,0,0	5,5,5
Bot: script typing at even speed	5,5,5	0,0,0	5,5,5

Strictness chosen: Balanced, Reason: Balanced blocked most impostors while rarely rejecting the real owner, whereas Strict rejected the owner too often and Relaxed let a friend with a similar typing style in. Testing was done on Different Laptops with 4-5 people, so these are small-sample results, not a full statistical evaluation.

7. Limitations and future work

- People with a similar typing style can sometimes pass; a stricter setting blocks them more often but also blocks the real owner more often.
- Tiredness, stress or a different keyboard can change someone's rhythm. Backspace, pasting and password-manager auto-fill are not supported.
- This is a browser demo: profiles are stored in localStorage and the decision runs in the page. A real system would store profiles and decide on a server, with salted password hashing, rate limiting and the live match bar hidden.
- A bot that adds random human-like variation could pass the bot rules, though it would still have to match the owner's rhythm. Future work: timing the username as well, mouse movement, touch input, and a stronger statistical model.

8. AI use

We used Claude (Anthropic) to help write the code. We tested the system ourselves, changed the design after our own test results (for example the Strictness setting, the 80% pass mark and 15 enrollment samples), and wrote up the results here.